

19 Poisson Distributions

- Binomial and Negative Binomial models have several restrictive assumptions that might not be satisfied in practice
- Poisson models are more flexible and are often used to model the distribution of random variables that count the number of “relatively rare” events that occur over a certain interval of time/in a certain location

Example 19.1. Recall the matching problem with a general n : there are n objects that are shuffled and placed uniformly at random in n spots with one object per spot. Note: n is a fixed and known number, but we are considering different values of it.

We have seen that $E(X) = 1$ for any n and that unless n is small (e.g., $n \leq 6$) the approximate distribution of X is the Poisson(1) distribution.

1. Starting with $x = 0$, describe the distribution of X in terms of relative likelihoods:

- 1 is [blank] times as likely as 0
- 2 is [blank] times as likely as 1
- 3 is [blank] times as likely as 2
- 4 is [blank] times as likely as 3
- 5 is [blank] times as likely as 4
- 6 is [blank] times as likely as 5
- In general, x is [blank] times as likely as $x - 1$, for $x = 1, 2, 3, \dots$

2. Specify the pmf of X .

3. Use the pmf to compute $E(X)$.

Table 19.1: Poisson(1) distribution, the approximate distribution of X , the number of matches in the matching problem for general n and uniformly random placement of objects in spots.

x	P(X=x)
0	0.3679
1	0.3679
2	0.1839
3	0.0613
4	0.0153
5	0.0031
6	0.0005
7	0.0001

4. Use the pmf to compute $\text{Var}(X)$.

- A random variable X has a **Poisson distribution** with parameter $\mu > 0$ if the possible values are $0, 1, 2, \dots$ and the pmf follows the

Poisson(μ) pattern: x is $\frac{\mu}{x}$ as likely as $x - 1$ (for $x = 1, 2, 3, \dots$)

- As a function of $x = 0, 1, 2, \dots$, a Poisson(μ) pmf increases for $x < \mu$ and then decreases for $x > \mu$
 - If μ is a whole number, then the Poisson pmf assigns the same probability to the values μ and $\mu - 1$.
 - If $\mu < 1$ then $x = 0$ has the highest probability and the pmf decreases from there.
- A discrete random variable X has a Poisson distribution with parameter $\mu > 0$ if and only if its probability mass function p_X satisfies

$$p_X(x) = \frac{e^{-\mu} \mu^x}{x!}, \quad x = 0, 1, 2, \dots$$

- The Poisson pattern implies that the shape of a Poisson pmf as a function of x is given by $\mu^x/x!$.

$$p(x) = p(0) \frac{\mu^x}{x!}, \quad x = 0, 1, 2, \dots$$

- The constant $p(0) = e^{-\mu}$ simply renormalizes the pmf so that the probabilities sum¹ to 1.

- If X has a Poisson(μ) distribution then

$$\begin{aligned} E(X) &= \mu \\ \text{Var}(X) &= \mu \end{aligned}$$

Example 19.2. Let X be the number of home runs hit (in total by both teams) in a randomly selected Major League Baseball game. Assume that X has a Poisson(2.4) distribution.

1. Interpret the parameter 2.4 in context.
2. Compute $P(X = 2.4)$.
3. Identify the most likely value of X .
4. Specify the pmf of X .
5. Compute $P(X = 0)$ and interpret the value in context.

¹Recall the series expansion $e^{\mu} = \sum_{x=0}^{\infty} \frac{\mu^x}{x!}$

6. Construct a table and spinner corresponding to the distribution of X .

Example 19.3. Suppose that the number of fatal crashes in SLO County in a week follows, approximately, a $\text{Poisson}(0.53)$ distribution, independently from week to week. Let X be the number of fatal crashes in a period of 4 consecutive weeks (close enough to a “month”).

1. What is $E(X)$? You should be able to compute it without knowing the distribution of X .
2. Compute $P(X = 0)$.
3. Compute $P(X = 1)$.
4. How could you use simulation to approximate the distribution of X ?
5. Use simulation to approximate the distribution of X . Does the approximate distribution follow (approximately) the Poisson pattern?

6. Use a Poisson pmf to compute $P(X = 5)$.

- **Poisson aggregation.** If X and Y are independent, X has a $\text{Poisson}(\mu_X)$ distribution, and Y has a $\text{Poisson}(\mu_Y)$ distribution, then $X + Y$ has a $\text{Poisson}(\mu_X + \mu_Y)$ distribution.
- If component counts are independent and each has a Poisson distribution, then the total count also has a Poisson distribution.